Pressurized Greenhouse: A Responsive Environment to Partial Gravity Conditions

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ABSTRACT

The greenhouse is a vital module for a long-duration mission to support fresh nutrition for the crew. This paper discusses two concepts of extraterrestrial greenhouses. The first one considers a habitable greenhouse, which has physical features of the habitat module. Human factors are the main focus of such design, aiming to design an interior garden for the crew. Since the priority is to support the crew's psychological health through interactions with plants, interior gardens are considered in both public and private areas, e.g., in the dining area or exercise module. The second approach is an industrial greenhouse to produce maximum nutrition with the lowest cargo mass and resource consumptions. Such a low-pressure greenhouse has the optimum environment for plant growth. The total interior pressure diminishes from the habitable zone of 10.2–14.7 psi (70.2–101.3 kPa) to the plant's living zone of 4.0 psi (27.6 kPa), and carbon dioxide level jumps from 400 ppm to 1500 ppm. Therefore, the crew would have to use EVA suits for operations within the greenhouse. This paper presents a new concept of a greenhouse as a prebreathing chamber because the pressure value does not affect the plant growth cycle. A greenhouse module would accommodate a spectrum of pressure, oxygen, and carbon dioxide levels from the plant's to human's living zones. To prevent decompression sickness, the crew has to dedicate 2–4 hours for prebreathing in an airlock before and after each EVA. From a psychological standpoint, spending time in a greenhouse environment would be a better alternative to airlock conditions. Besides, exercising by the plants, gardening, and/or relaxing after an EVA would be more beneficial for the crew comparing to spending hours in a tight airlock. Moreover, the environmental control and life support system (ECLSS) of the greenhouse remains independent from the habitat and in case of any emergencies can function separately and help the crew to survive. In the long term, a lower level of oxygen reduces oxidizing of the interior of the module. In summary, this paper presents a responsive design concept for the described interior transitions. The proposed design is evaluated using figures of merits (FOM) by the level of their impact on the overall mission planning and success.

INTRODUCTION

For humans to survive a long-duration mission on Mars's surface, the presence of a greenhouse module is essential. The crew cannot bring all the required food supplies with them from Earth and would have to grow food on Mars. To date, two categories of greenhouse designs have been proposed: an industrial-scale greenhouse, which prioritizes efficiency partly through maintaining a low-pressure environment, and a habitable greenhouse, which supports psychological wellness by maintaining a fully pressurized environment that crewmembers can

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move freely in. This paper presents a hybrid approach in which a greenhouse module could support a variety of pressures, allowing for benefits from both types of designs and providing other advantages as well.

HABITABLE GREENHOUSES

In the assumptions of NASA's Big Idea Challenge Competition 2018 (*NASA's BIG Idea Challenge*, 2018) and other recent projects (Rivas et al., 2019)(Hinterman et al., 2019)(Bader et al., 2019)(Hava et al., 2019)(Birtell et al., 2019). It is assumed that the greenhouse connects to the habitat after crew arrival and integrates into the logistics/service module. In this scenario, the habitat connects to the greenhouse directly or through an inflatable tunnel allowing the crew to access the greenhouse at any time.

This approach assumes that the habitat and greenhouse utilize a similar structure and function under equal pressure (Morris et al., 2016). That simplifies an overall system and enables modules' individual ECLSS systems to unify as one expanded system. Since there is no airlock in between, all atmospheric parameters such as pressure, temperature, humidity, etc. would be the same in all modules and similar to earth conditions.

Psychologically, the ease of access to the greenhouse creates a feeling of it as a backyard or a garden. In the Martian Integrated Nourishment Aid (MINA) project (Moghimi Esfandabadi & Bannova, 2019b), we proposed a greenhouse with a private area for each crewmember to enhance living conditions on Mars with improved design and culturally compatible food selections.

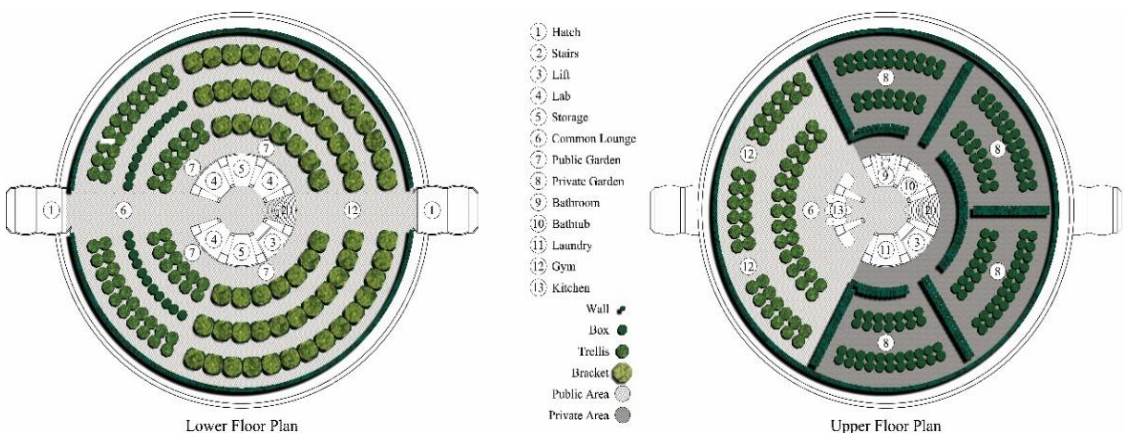

Figure 1 Floor Plans of MINA(Moghimi Esfandabadi & Bannova, 2019b)

INDUSTRIAL GREENHOUSE APPROACH

A traditional approach to industrial greenhouse design aims to maximize crop yields and daily calorie intake while producing fewer plants. A cultural aspect of the food, in this case, is neglected, and cuisine variety or food customization is not considered (Moghimi Esfandabadi & Bannova, 2019a). Such an approach ignores physiological and psychological human factors associated with food preparation and consumption behavior.

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Figure 2 Public Garden (Moghimi Esfandabadi & Bannova, 2019b)

Figure 3 Private Garden(Moghimi Esfandabadi & Bannova, 2019b)

These greenhouses are designed to maintain a low-pressure environment to minimize the use of resources. The low equivalent system mass of such greenhouse systems makes for reduced total launch mass (Ferl et al., 2002). Figure 4 demonstrates the contributions of mass, power, and thermal insulation to the equivalent system mass of a greenhouse assembly. A pressure under 60 kPa allows for using new types of thermal insulation, such as materials based on SiO_2 aerogel (Liu et al., 2016) or their combination with vacuum insulation panels (Thapliyal & Singh, 2014). The application of aerogel-type materials can provide translucent qualities of the external envelop structure (Saboktakin & Saboktakin, 2015). Therefore, greenhouse lighting may be natural, artificial, or hybrid. The comparison between different types of lighting in Figure 4 demonstrates that changing artificial to natural/hybrid lighting would reduce the mass of the inflatable by 32.5% compared to a high-pressure greenhouse (59.2 kPa) and 39.9% compared to a low-pressure greenhouse (30.0 kPa). The overall mass of a low-pressure greenhouse shell using natural/hybrid lighting is only 50.1% of that of a high-pressure greenhouse using artificial lighting (Hublitz et al., 2004).

Moreover, lowering the internal pressure reduces the consumption of resources, mainly O_2 and N_2, leading to faster initiation of operations. Hybrid lighting of such a structure reduces power requirements in comparison to the total artificial lighting of a higher-pressure greenhouse.

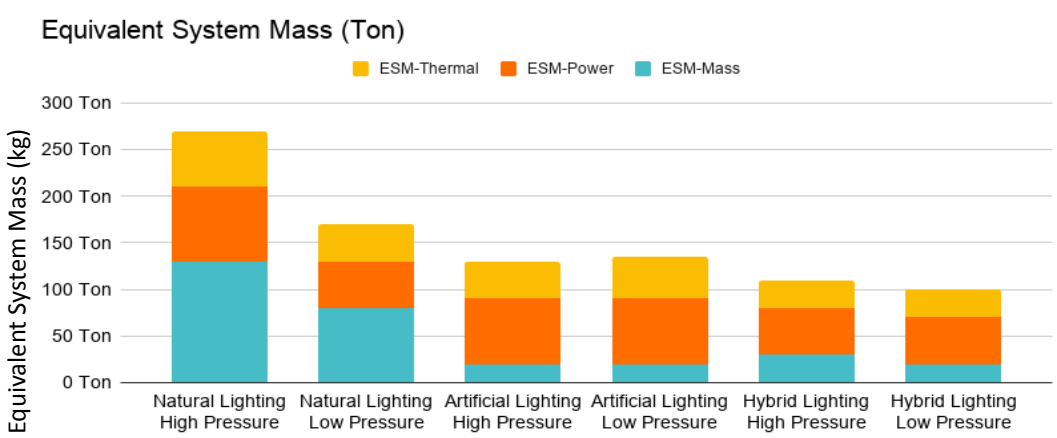

Figure 4 Contribution of mass, power and thermal to Equivalent System Mass of a greenhouse assembly (Hublitz et al., 2004)

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Higher plants tolerate hypobaria at least down to 10 kPa (Brinkjans, 1996) (Massimino & André, 1999) (Goto et al., 2011). With the earth-like ambient atmospheric gas composition, reduced pressure decreases plant growth, increases their respiration, causes stomatal closure, and induces expression of multiple genes mostly related to drought and/or anoxia (lack of oxygen) stress (Paul et al., 2004). However, reducing the ratio of O_2/CO_2 and increasing the partial pressure of CO_2 could reverse this effect (Corey et al., 1997)(Miśkiewicz et al., 2000). It should be noted that at all atmospheric conditions, a minimum of 2% of O_2 and a maximum of 0.2% of CO_2 should be maintained (Lehto et al., 2006). Inert gases like N_2 or He_2 supply the remaining 98% to stabilize the pressure on a total of 10 kPa. Low atmospheric pressure, though, reduces the conductive and convective cooling/heating performance. Therefore, low-pressure greenhouses require an adequate ventilation system to maintain gas composition.

THE MIDDLE PATH

A third greenhouse design approach is to allow its internal pressure to fluctuate, accommodating the needs of crewmembers and plants alike. During crewed EVA surface missions, the environment can adapt to human physiological needs; meanwhile, without the crew, the greenhouse can continue working at its lowest mode.

The Pre-Breathe Greenhouse

The major problem of movement from the atmospheric pressure (101.3 kPa) of the cabin to lower pressures is the threat of aerospace bends-decompression sickness. (Nitrogen gas comes out of solution in the bloodstream and forming bubbles that collect painfully in the joints of the body). To avoid this problem, astronauts "pre-breathe" pure oxygen for at least three hours before donning their suits to purge the nitrogen from their bloodstream (Cohen, 2000). Table 1 shows different pressures and atmospheric compositions used in space missions. A pre-breathe time of 1 hour is assumed as a tentative upper bound for surface exploration EVAs to denitrogenate the brain and spinal cord and prevent (Type II) DCS symptoms (Anderson et al., 2015). Long pre-breathing periods diminish productive crew time and maybe unacceptable during lunar or Mars missions when EVAs will be more frequent and should be easy, simple, routine, and safe.

For surface missions, the pressure in a Multi-Mission Space Exploration Vehicle would be lowered from the normal 101.3 kPa of the habitat to 56.5 kPa to minimize the rate of nitrogen return while breathing cabin air - a nitrogen/oxygen mixture. This technique allows for a short 30-40 minute pre-breathe time in the Extravehicular Mobility Unit just before starting an EVA (Newman & Barratt, 1997).

There are four different protocols for surface EVAs. Table 2 displays that protocols A and B are straight 8 hours of EVA, and protocols C and D have two hours duration. While protocols A and C have 80% of oxygen, the B and D protocols have 95%. These protocols show the depressurization process from the assumed cabin pressure of 56.5 kPA to 29.7 kPa of the spacesuit, which takes 25 minutes. In the protocols C and D, there is one hour of the hold before starting each EVA that increases each pre-breathe cycle to one hour and 25 minutes and a total of 4 hours and 15 minutes.

In the current mission scenarios, the crew would spend the pre-breathe time in an airlock module that is equipped with gym facilities. A proposed low-pressure greenhouse can enhance the crew's pre-breathing routine and reduce stress levels by offering interactions with plants or doing daily agricultural chores. Also, exercising around plants is beneficial for the plants, the ECLSS system, and the crew.

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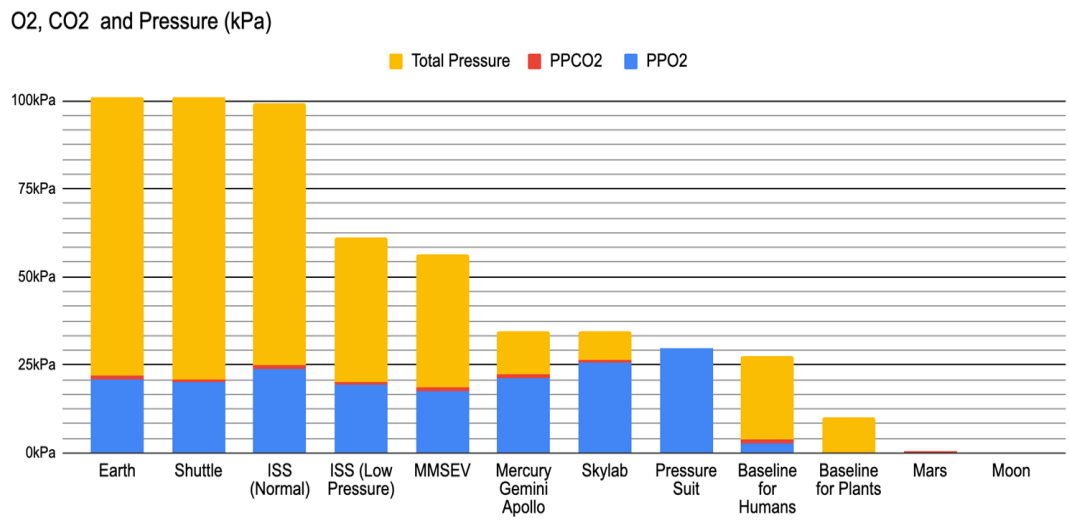

Figure 5 Pressure and Partial Pressure in Different Programs

Table 1 EVA protocols with different combinations of IR and O_2 breathing mixtures using the new exploration atmosphere. In all cases, the balance of the gas is N_2 (Abercromby et al., 2015)

EVA protocols with different combinations of IR and O_2 breathing mixtures using the new exploration atmosphere. In all cases the balance of the gas is N_2.

Protocol A: Continuous EVA, 80% O_2					Protocol B: Continuous EVA, 95% O_2				
Step		Duration min	Pressure kPa (psi)	O_2 %	Step		Duration min	Pressure kPa (psi)	O_2 %
1	Saturated		56.5 (8.2)	34	1	Saturated		56.5 (8.2)	34
2	Purge	0:02	56.5 (8.2)	80	2	Purge	0:08	56.5 (8.2)	95
3	Depress	0:01	56.5→41.4 (8.2→6.0)	80	3	Depress	0:01	56.5→41.4 (8.2→6.0)	95
4	Prebreathe[a]	0:15	41.4 (6.0)	80	4	Prebreathe[a]	0:15	41.4 (6.0)	95
5	Depress	0:01	41.4→29.7 (6.0→4.3)	80	5	Depress	0:01	41.4→29.7 (6.0→4.3)	95
6	EVA	8:00[b]	29.7 (4.3)	80	6	EVA	8:00[b]	29.7 (4.3)	95
Protocol C: 3 × 2 h EVAs, 80% O_2					Protocol D: 3 × 2 h EVAs, 95% O_2				
Step		Duration min	Pressure kPa(psi)	O_2 %	Step		Duration min	Pressure kPa (psi)	O_2 %
1	Saturated		56.5 (8.2)	34	1	Saturated		56.5 (8.2)	34
2	Purge	0:02	56.5 (8.2)	80	2	Purge	0:08	56.5 (8.2)	95
3	Depress	0:01	56.5→41.4 (8.2→6.0)	80	3	Depress	0:01	56.5→41.4 (8.2→6.0)	95
4	Prebreathe[a]	0:15	41.4 (6.0)	80	4	Prebreathe[a]	0:15	41.4 (6.0)	95
5	Depress	0:01	41.4→29.7 (6.0→4.3)	80	5	Depress	0:01	41.4→29.7 (6.0→4.3)	95
6	EVA	2:00	29.7 (4.3)	80	6	EVA	2:00	29.7 (4.3)	95
7	Repress	0:02	29.7→56.5 (4.3→8.2)	34	7	Repress	0:02	29.7→56.5 (4.3→8.2)	34
8	Hold	1:00	56.5 (8.2)	34	8	Hold	1:00	56.5 (8.2)	34
	Repeat steps 2–8 twice, for 3 total EVAs					Repeat steps 2–8 twice, for 3 total EVAs			

[a] EVA can begin at 6 psi at the start of prebreathe, with suit pressure being dropped to 29.7 kPa (4.3 psi) 15 min after start of prebreathe clock.
[b] $P(DCS)$ was calculated after 6 h and after 8 h of continuous EVA.

The presented in this paper pre-breathe greenhouse architecture utilizes 34.48 to 56.5 kPa pressure to accommodate two operational modes described in figures 6 and 7. The time spent in the greenhouse will vary depending in the operational requirements and can be either 8 hours or 3 incremental periods of 2 hours each (Table 2).

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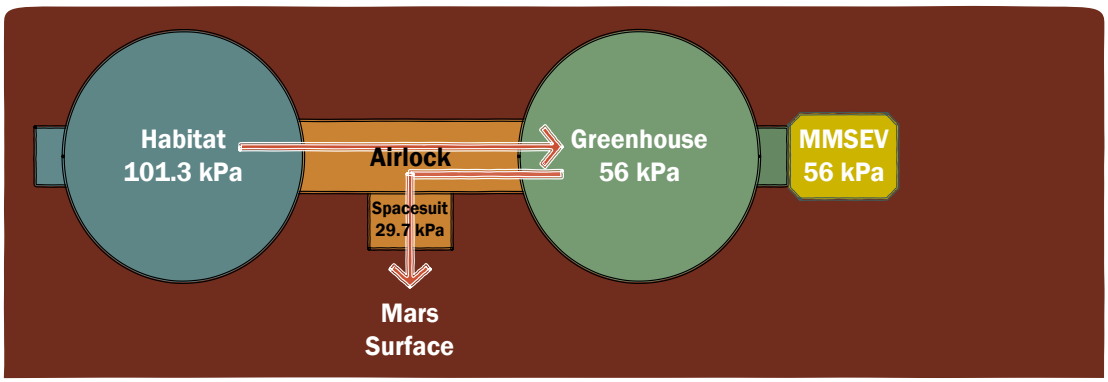

Figure 6 Diagram of proposed pre-breathe for spacesuit surface mission

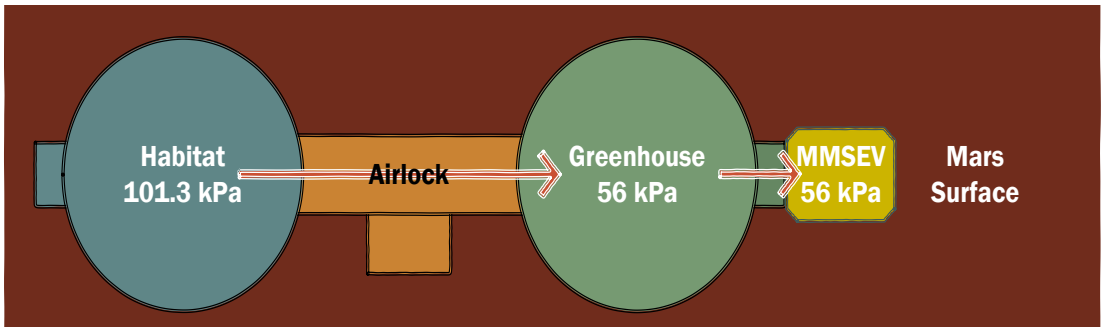

Figure 7 Diagram of proposed pre-breathe for MMSEV surface mission

Table 2 Total Spent Time in Greenhouse

Scenario 3x2h																Duration
Time	06:00	08:00	08:20	10:20	11:20	11:40	13:40	14:40	15:00	16:20	16:40	17:00	17:20	18:40	19:20	
Location	Pre-breathe			EVA1	Hold	Pre-breathe	EVA2	Hold	Pre-breathe	EVA3			Pre-breathe			7:20
Pressure	101.3	56.5	29.7	29.7	56.5	56.5	29.7	56.5	56.5	29.7	29.7	29.7	29.7	56.5	101.3	
Scenario 1x8h																
Location	Pre-breathe												Pre-breathe			
Pressure	101.3	56.5	29.7	29.7	29.7	29.7	29.7	29.7	29.7	29.7	29.7	29.7	56.5	101.3		4:40

SELECTION OF VALIDATION CRITERIA

This paper discussed possible greenhouse designs utilizing different geometries of inflatable modules but with the same overall interior area and volume. The selection of a greenhouse design approach depends on mission type, objectives, and surface capabilities. The most mission-appropriate greenhouse design solutions have to be selected by the level of their impact on the overall mission planning and success. Evaluation criteria or Figures of Merits (FOM) include the type of mission and mission duration, resources requirements and availability of in-situ resources, number of precursor missions and pre-deployed surface elements, number of crew, readiness levels of proposed technologies, greenhouse environmental impact on habitat systems and structures (Table 3).

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Table 3 Figure of Merit

FOM	Private Garden	Industrial Greenhouse	Pre-Breathe Greenhouse
# Crew	4	4 Or More	4 Or More
Nutrition Support	Low/Medium	High	High
Mission Duration	Short	Long/Settlement	Long
Resources	High	High	Medium
Autonomous Operations	Low	High	Low/High
Functionality	Single Function	Single Function	Multi-Functional
Impact on Habitat Systems	High	Low	Medium/Low
Psychological Benefits	High	Low/Medium	High/Medium
Area/Volume Utilization	Not Efficient	Max Efficient	Efficient

CONCLUSIONS

A pre-breathe greenhouse approach may offer the maximum benefits while utilizing fewer resources than other described in this paper greenhouse architectures offer. A private garden within a habitat and industrial-style greenhouses either add complexity and load on habitat systems and/or require massive construction resources and efforts and delay operational readiness of the greenhouse itself. A compromise between the two is a pre-breather architecture that offer multifunctional capabilities, low impact on habitat systems and structures, and lower launch/landing mass requirements, while providing 24/7 availability for the crew.

REFERENCES

Abercromby, A. F. J., Conkin, J., and Gernhardt, M. L. (2015). Modeling a 15-min extravehicular activity pre-breathe protocol using NASAs exploration atmosphere (56.5 kPa/34% O2). *Acta Astronautica*, *109*, 76–87. https://doi.org/10.1016/j.actaastro.2014.11.039

Anderson, M. S., Ewert, M. K., Keener, J. F., and Wagner, S. A. (2015). *Life Support Baseline Values and Assumptions Document*. March 2015.

Bader, D., Brown, L., Byland, J., Chamberlin, A., Elliot, I., George, C., Juarez, N., and Liao, J. (2019). *Martian Agriculture and Plant Science Greenhouse (MAPS)*.

Birtell, E., Bunchek, J., Greaves, B., Onsay, D., and Torres, J. (2019). *GAIA (Greenhouse Attachment for the Ice Home Architecture)*.

Brinkjans, H. J. (1996). *Plant responses to reduced air pressure : advanced techniques and results*. *18*(4). https://doi.org/10.1016/0273-1177(95)00889-m

Cohen, M. M. (2000). Pressurized Rover Airlocks. *Engineering*, *724*. https://doi.org/10.4271/2000-01-2389

Corey, K. A., Barta, D. J., and Henninger, D. L. (1997). Photosynthesis and respiration of wheat stand at reduced atmospheric pressure and reduced oxygen. *Advances in Space Research*, *20*(10), 1869–1877. https://doi.org/10.1016/S0273-1177(97)00854-5

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Ferl, R. J., Schuerger, A. C., Paul, L.-A., Gurley, W. B., Corey, K. A., and Bucklin, R. A. (2002). Plant Adaptation To Low Atmospheric Pressures : Potential Molecular Responses. *Apollo The International Magazine Of Art And Antiques*, *352*, 1–17.

Goto, E., Ohta, H., Iwabuchi, K., and Takakura, T. (2011). Measurement of Net Photosynthetic and Transpiration Rates of Spinach and Maize Plants under Hypobaric Condition. *Journal of Agricultural Meteorology*, *52*(2), 117–123. https://doi.org/10.2480/agrmet.52.117

Hava, H., Zhou, L., and Lombardi, E. M. (2019). *SIRONA: Sustainable Integration of Regenerative Outer - Space Nature and Agriculture*.

Hinterman, E., Stamler, N., Barber, S., Nowak, H., Zhan, Z., Kusters, J., Seaman, S., and Schneiderman, T. (2019). *Biosphere Engineered Architecture for Viable Extraterrestrial Residence (BEAVER)*.

Hublitz, I., Henninger, D. L., Drake, B. G., and Eckart, P. (2004). Engineering concepts for inflatable Mars surface greenhouses. *Advances in Space Research*, *34*(7 SPEC. ISS.), 1546–1551. https://doi.org/10.1016/j.asr.2004.06.002

Lehto, K. M., Lehto, H. J., and Kanervo, E. A. (2006). Suitability of different photosynthetic organisms for an extraterrestrial biological life support system. *Research in Microbiology*, *157*(1), 69–76. https://doi.org/10.1016/j.resmic.2005.07.011

Liu, Z. Hui, Wang, F., and Deng, Z. Ping. (2016). Thermal insulation material based on SiO2 aerogel. *Construction and Building Materials*, *122*, 548–555. https://doi.org/10.1016/j.conbuildmat.2016.06.096

Massimino, D., and André, M. (1999). Growth of wheat under one-tenth of the atmospheric pressure. *Advances in Space Research*, *24*(3), 293–296. https://doi.org/10.1016/S0273-1177(99)00316-6

Miśkiewicz, E., Ivanov, A. G., Williams, J. P., Khan, M. U., Falk, S., and Huner, N. P. A. (2000). Photosynthetic acclimation of the filamentous cyanobacterium, Plectonema boryanum UTEX 485, to temperature and light. *Plant and Cell Physiology*, *41*(6), 767–775. https://doi.org/10.1093/pcp/41.6.767

Moghimi Esfandabadi, M., and Bannova, O. (2019a). Designing A Martian Greenhouse as A Habitable Space: Feasibility Studies and Design Approach. *70th International Astronautical Congress (IAC)*.

Moghimi Esfandabadi, M., and Bannova, O. (2019b). Martian Greenhouse Architecture: Enabling Habitability, Safety, and Aesthetics. *70th International Astronautical Congress (IAC)*.

Morris, M., Ciardullo, C., Lents, K., Montes, J., Rudakevych, O., Sono, M., Sono, Y., and Yashar, M. (2016). Mars Ice House: Using the Physics of Phase Change in 3D Printing a Habitat with H2O. *AIAA SPACE* 2016, September, 1–15. https://doi.org/10.2514/6.2016-5528

NASA's BIG Idea Challenge. (2018). NASA. http://bigidea.nianet.org/competition-basics/

Newman, D., and Barratt, M. (1997). Life Support and Performance Issues for Extravehicular Activity (EVA). In *Fundamentals of Space Life Sciences* (Vol. 66, pp. 37–39). Krieger, Melbourne, FL. https://doi.org/https://doi.org/10.4271/951592

Paul, A. L., Schuerger, A. C., Popp, M. P., Richards, J. T., Manak, M. S., and Ferl, R. J. (2004). Hypobaric Biology: Arabidopsis Gene Expression at Low Atmospheric Pressure. *Plant Physiology*, *134*(1), 215–223. https://doi.org/10.1104/pp.103.032607

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Rivas, Z. A., Escalona, A., Genszler, G., Hodsden, T. L., Mcgonagle, M., Dick, D., Mahoney, P., Yu, C., Cushman-roisin, B., Ray, L., Collier, J., and Lynd, L. (2019). *DEMETER Advanced Concept of Operations*.

Saboktakin, A., and Saboktakin, M. R. (2015). Improvements of reinforced silica aerogel nanocomposites thermal properties for architecture applications. *International Journal of Biological Macromolecules*, *72*, 230–234. https://doi.org/10.1016/j.ijbiomac.2014.08.024

Thapliyal, P. C., and Singh, K. (2014). Aerogels as Promising Thermal Insulating Materials: An Overview. *Journal of Materials*, *2014*, 1–10. https://doi.org/10.1155/2014/127049

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